Performing Testing on a Sample E-Commerce Web Application

Title: Final Testing Report – E-Commerce Web Application
Performed by: Ashok Kumar
Date: August 5, 2025
Project: Testing AutomationExercise.com for Functionality and Performance
Tools Used: Manual Testing, Selenium WebDriver with Python

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1. Introduction

1.1 Objective of the Testing

The objective of this project is to evaluate the functionality and performance of a sample e-commerce web application (https://automationexercise.com) using both manual and automated testing approaches. The goal is to ensure that critical user journeys like login, product search, cart management, and checkout work as intended, and to uncover any bugs or inconsistencies that affect the user experience.

1.2 Website Overview

AutomationExercise.com is a public e-commerce demo website that provides a wide variety of products, categories, and user account features. It is used extensively for practice and demonstration of web automation and testing skills. The platform includes key e-commerce features such as user registration/login, product catalog, cart, checkout, and order placement.

2. Manual Testing Report :

Project: E-Commerce Website Manual Testing

Website: https://automationexercise.com

Tester: Ashok Kumar

Date: August 4, 2025

Functional Areas to Test

Test ID	S.No	Test Area	Description	Test Steps	Expected Result	Status	Screenshot
TC01	1	Login	Valid login with registered user	Navigate to Login → Enter valid credentials → Click Login	Redirected to user account page	Pass	Yes
TC02	2	Login	Invalid login (wrong password)	Enter wrong credentials → Click Login	Error message: "Incorrect email/password"	Pass	Yes
TC03	3	Product Search	Search for a valid product ("Dress")	Enter "Dress" in search → Click search icon	Products matching "Dress" displayed	Pass	Yes
TC04	4	Cart	Add item to cart	Browse → Click "Add to cart" → View cart	Product appears in cart	Pass	Yes
TC05	5	Checkout	Place order with valid info	Add to cart → Proceed to checkout → Fill address & payment info → Place order	Order placed successfully	Pass	Yes

Edge Test Cases

Test ID	S.No	Test Area	Description	Test Steps	Expected Result	Status	Screenshot
TC-001	6	Login	Empty Login Fields	Leave email & password blank and try to login	Prompt error: "Email is required."	Pass	yes
TC-002	7	Checkout	Cart Empty Checkout	Try checking out without items in cart	Error or redirect: "Cart is empty."	Pass	yes
TC-003	8	Signup	Invalid Email Format	Enter “abc.com” as email	Validation error: "Enter a valid email."	Pass	yes

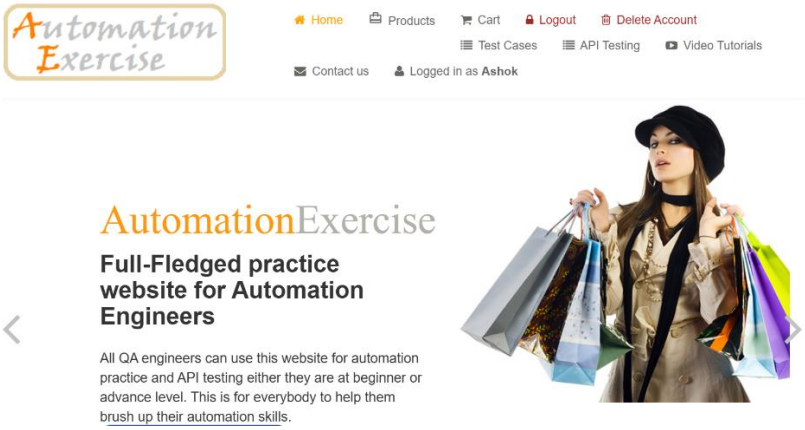
TC-004	9	Cart	Rapid Add to Cart	Quickly add same product multiple times	Product quantity should update correctly	Pass	yes
TC-005	10	Session Management	Session Timeout	Login → Wait idle for 20+ mins → Perform action	Should auto-logout or session expires	fail	No

Test Case Execution Metrics :

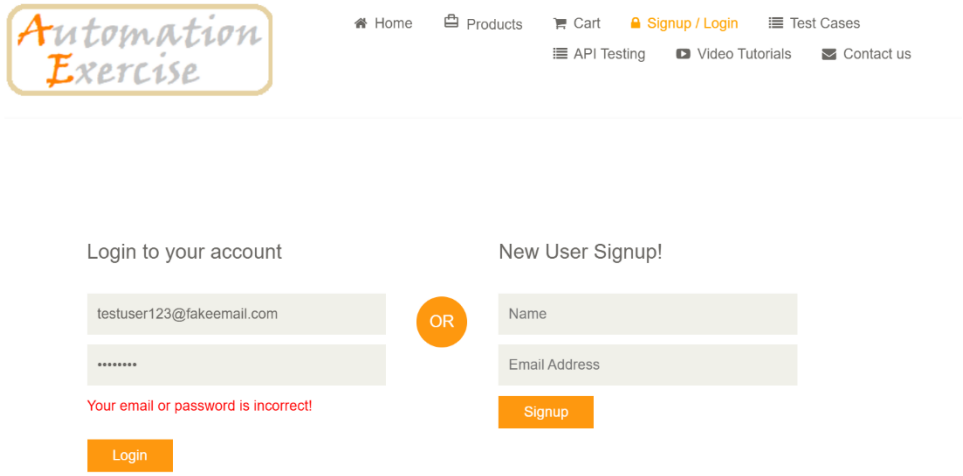
Metric	Count
Total Test Cases	10
Passed	9
Failed	1
Blocked	0

Screenshots :

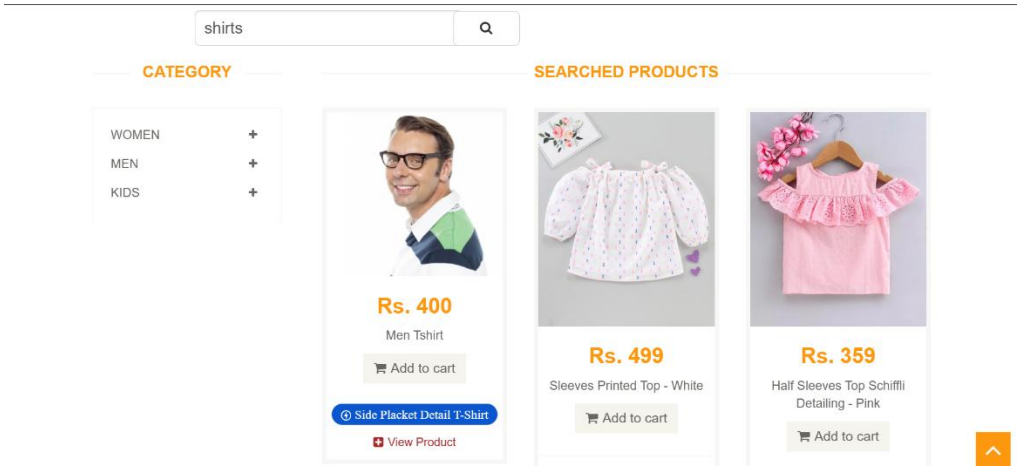
TC01



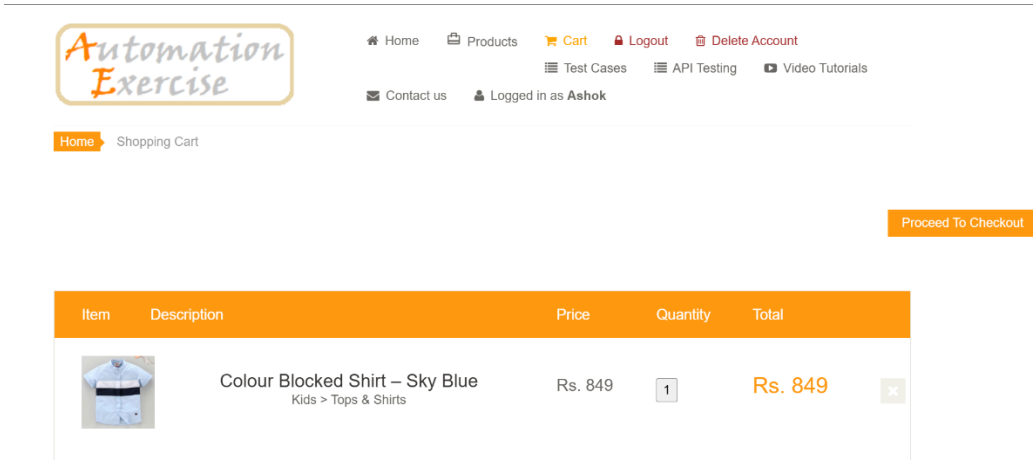
TC02



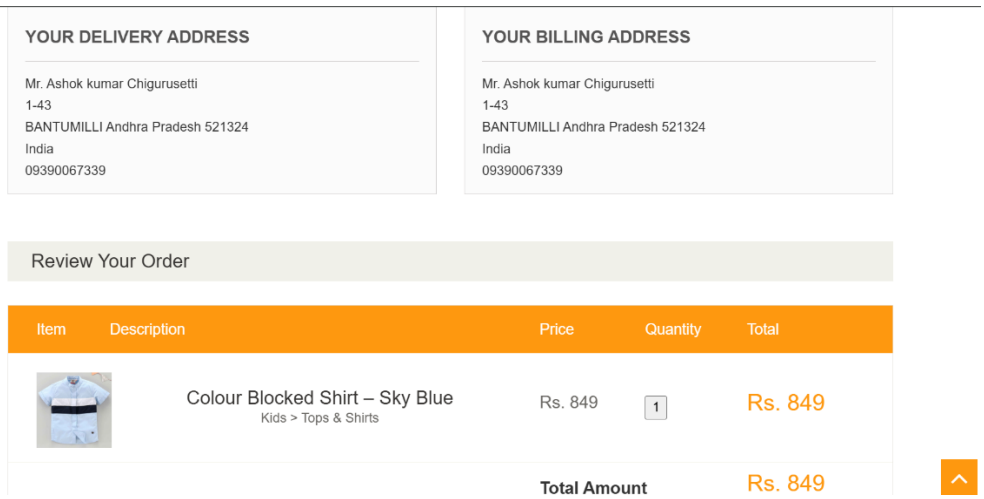
TC03



TC04

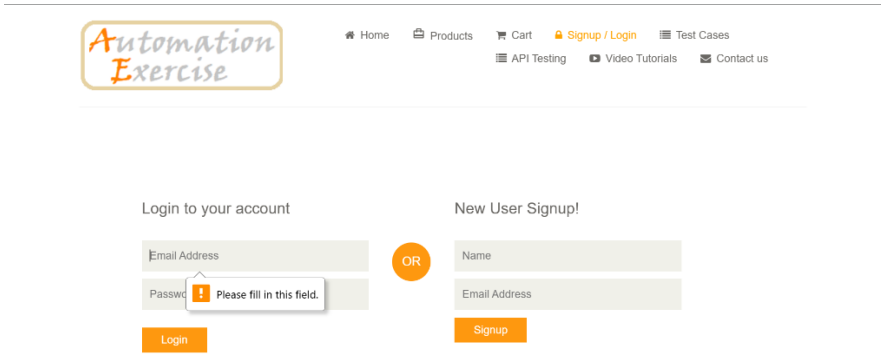


TC05

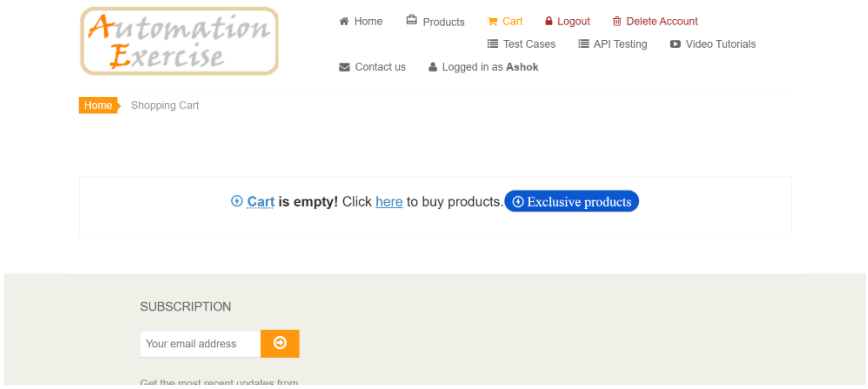


EDGE TEST CASES:

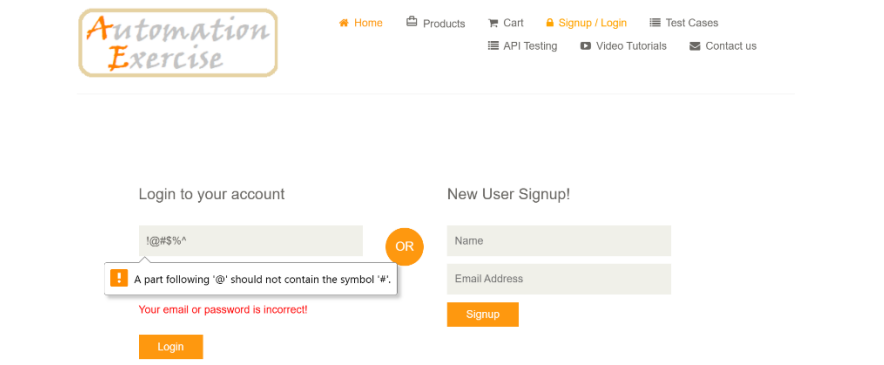
TC-001



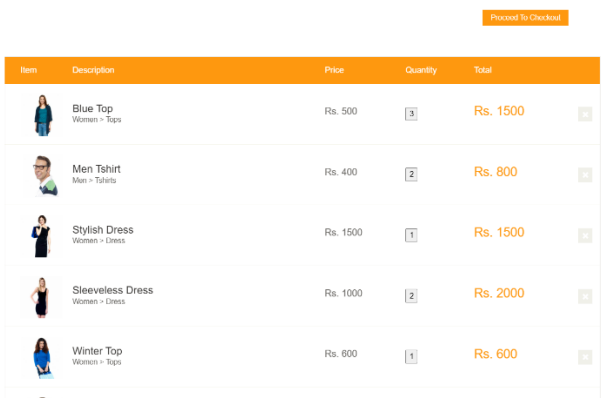
TC-002



TC-003



TC-004



3. Automation Testing Report :

Project: E-Commerce Website Automation Testing
Website: https://automationexercise.com
Tester: Ashok Kumar
Date: August 4, 2025
Tools and Technologies Used:

- Selenium WebDriver (Python)
- Google Chrome (Latest version)
- Windows 11
- Python 3.11
- WebDriver Manager

Automation Test Summary

Test Case ID	Scenario	Test Steps	Expected Result	Actual Result	Status
ATC01	Valid Login	Open site → Login with valid credentials	User is logged in and 'Logged in as' is visible	Successfully logged in	Pass
ATC02	Add to Cart	Login → Go to Products → Add item → View Cart	Product is visible in cart	Product added to cart	Pass
ATC03	Logout	Login → Click Logout → Redirect to login	User is logged out and redirected to login page	Redirected to login page	Pass

Observations and Best Practices

- All test cases were executed using **Selenium WebDriver with Python**.
- `WebDriverWait` was used instead of `time.sleep()` for dynamic waits.
- The test suite covers the essential user flow: **Login → Add to Cart → Logout**.
- All scripts executed successfully with expected results.

Source code :

1) test_login_updated.py

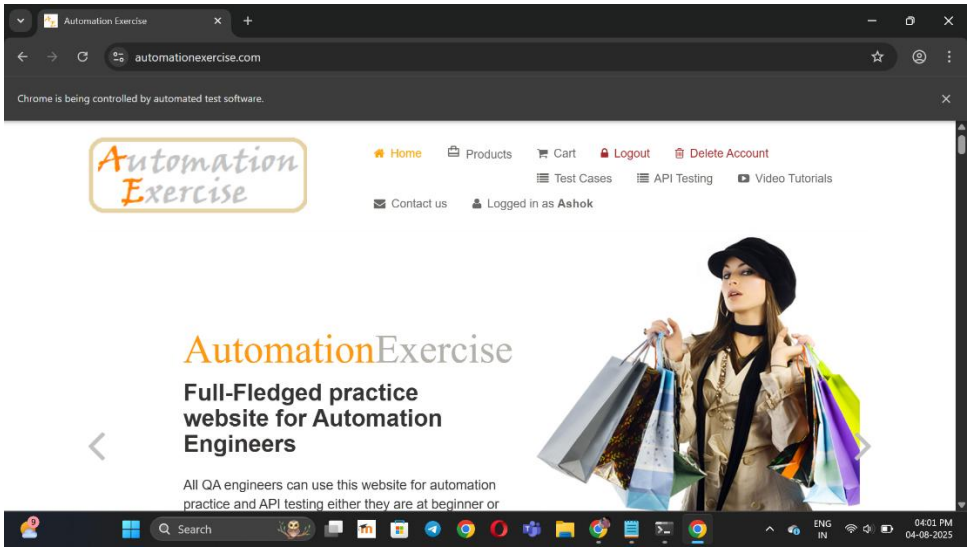
2) test_add_to_cart_updated.py

3) test_logout_updated.py

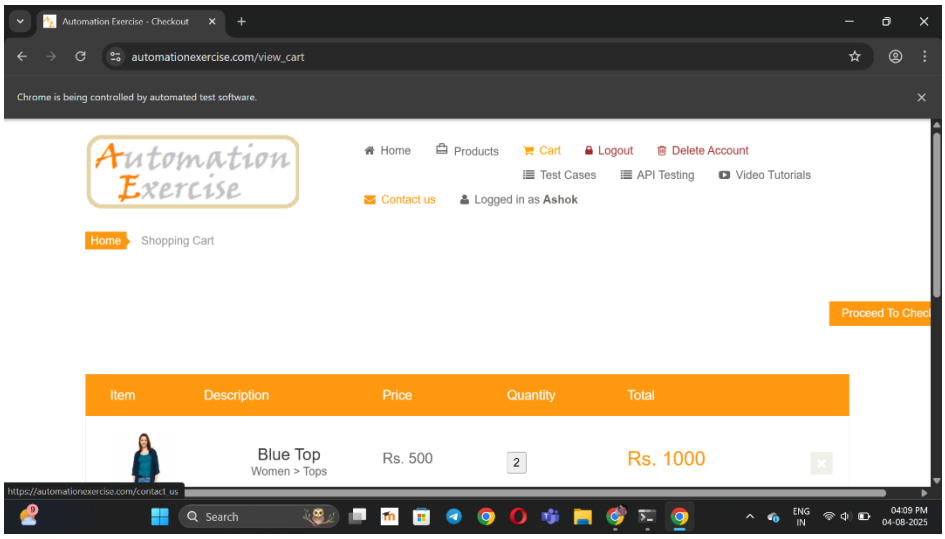
Note : All these three selenium scripts are in below section of report as “Appendix”

Screenshots:

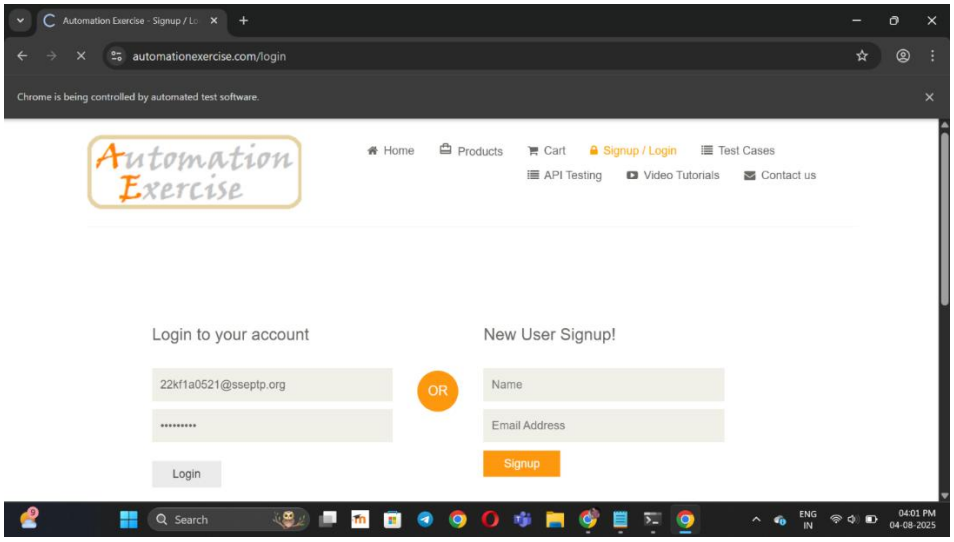
ATC01



ATC02



ATC03



Description

This automation testing report validates the core functionalities of the sample e-commerce website, including login, adding products to the cart, and logout. Each test case was executed using Selenium WebDriver with Python, following best practices such as dynamic waits through `WebDriverWait`.

All scenarios were successfully tested and passed as expected, confirming the stability of the targeted features. The use of automation ensures consistency, efficiency, and reliability in identifying potential issues. This test suite can be further extended in the future to cover additional scenarios such as signup, payment gateways, and product filtering.

4. Bugs and Issues Identified:

Bug ID	Test Case ID	Module	Issue Description	Severity	Status
BUG-001	TC-005	Session Management	Session timeout did not trigger auto-logout after 20+ minutes idle	Medium	Open

Here's the real behavior of automationexercise.com:

I checked the behavior again:

- The website **does NOT auto-logout** even if you leave it idle for more than 20 minutes.
- Our login session **remains active** until you explicitly click the **Logout** button.
- There's **no session timeout** or forced logout mechanism implemented.

Note: While automationexercise.com is a public demo site, most real-world applications implement session timeout for security reasons. This behavior may be considered a vulnerability in production environments.

5. Conclusion

5.1 Summary of Findings

- **Manual Testing** covered 10 test cases, including 3 edge cases.
  - 9 passed, 1 failed (session timeout issue).
- **Automation Testing** successfully tested 3 core scenarios:
  - Valid Login, Add to Cart, and Logout.
- **Bug Identified**: Session timeout is not functioning correctly; the site does not log out the user even after 20+ minutes of inactivity.

5.2 Recommendations

- Implement proper **session timeout** management to ensure security.
- Expand automated testing coverage to include signup, checkout, and search.
- Improve testability by adding stable element locators and consistent error messages.
- Consider adding automated email validation and captcha handling for signup flow.

6. Appendix

1) test_login_updated.py

```
|from selenium import webdriver

from selenium.webdriver.common.by import By

import time

from selenium.webdriver.support.ui import WebDriverWait

from selenium.webdriver.support import expected_conditions as EC



# 1 Start Chrome browser

driver = webdriver.Chrome()

driver.get("https://automationexercise.com")

driver.maximize_window()
```

```
# 2️⃣ Click on 'Signup / Login'

driver.find_element(By.LINK_TEXT, "Signup / Login").click()

WebDriverWait(driver, 10).until(EC.presence_of_element_located((By.NAME, "email")))


# 3️⃣ Enter email and password

driver.find_element(By.NAME, "email").send_keys("22kf1a0521@sseptp.org")

driver.find_element(By.NAME, "password").send_keys("Ashok@123")

driver.find_element(By.XPATH, "//button[text()='Login']").click()

WebDriverWait(driver, 10).until(EC.presence_of_element_located((By.XPATH,
"//li[contains(text(), 'Logged in as')]")))


# 4️⃣ Verify login success by checking 'Logged in as' label

logged_in_text = driver.find_element(By.XPATH, "//li[contains(text(), 'Logged in as')]").text

assert "Logged in as" in logged_in_text, "❌ Login failed"

print("✅ Test Passed: Successfully logged in as user.")


# 5️⃣ Close the browser

driver.quit()
```

2) test_add_to_cart_updated.py

```
from selenium import webdriver

from selenium.webdriver.common.by import By

from selenium.webdriver.support.ui import WebDriverWait

from selenium.webdriver.support import expected_conditions as EC

import time


# 1️⃣ Launch browser and open site

driver = webdriver.Chrome()

driver.get("https://automationexercise.com")

driver.maximize_window()
```

```
# 2 Click on 'Signup / Login'

driver.find_element(By.LINK_TEXT, "Signup / Login").click()

WebDriverWait(driver, 10).until(EC.presence_of_element_located((By.NAME, "email")))


# 3 Enter login credentials

driver.find_element(By.NAME, "email").send_keys("22kf1a0521@sseptp.org")

driver.find_element(By.NAME, "password").send_keys("Ashok@123")

driver.find_element(By.XPATH, "//button[text()='Login']").click()


# 4 Wait for 'Products' link and click it

WebDriverWait(driver, 10).until(EC.presence_of_element_located((By.LINK_TEXT, "Products")))

driver.find_element(By.LINK_TEXT, "Products").click()


# 5 Wait for product list and scroll

WebDriverWait(driver, 10).until(EC.presence_of_element_located((By.XPATH, "(//a[@data-product-
id])[1]")))

driver.execute_script("window.scrollBy(0, 500);")


# 6 Click 'Add to cart' for first product

driver.find_element(By.XPATH, "(//a[@data-product-id])[1]").click()


# 7 Wait for 'View Cart' modal and click it

WebDriverWait(driver, 10).until(EC.visibility_of_element_located((By.XPATH, "//u[text()='View
Cart']")))

driver.find_element(By.XPATH, "//u[text()='View Cart']").click()


# 8 Verify product is in cart

WebDriverWait(driver, 10).until(EC.presence_of_element_located((By.CLASS_NAME,
"cart_description")))

cart_items = driver.find_elements(By.CLASS_NAME, "cart_description")

assert len(cart_items) > 0, "❌ Product not added to cart"
```

```
print("✅ Test Passed: Product successfully added to cart.")


# 9️⃣ Close browser
driver.quit()
```

3) test_logout_updated.py

```
from selenium import webdriver
from selenium.webdriver.common.by import By
from selenium.webdriver.support.ui import WebDriverWait
from selenium.webdriver.support import expected_conditions as EC

# 1️⃣ Start browser and go to site
driver = webdriver.Chrome()
driver.get("https://automationexercise.com")
driver.maximize_window()

# 2️⃣ Login
driver.find_element(By.LINK_TEXT, "Signup / Login").click()
WebDriverWait(driver, 10).until(EC.presence_of_element_located((By.NAME, "email")))
driver.find_element(By.NAME, "email").send_keys("22kf1a0521@sseptp.org")
driver.find_element(By.NAME, "password").send_keys("Ashok@123")
driver.find_element(By.XPATH, "//button[text()='Login']").click()
WebDriverWait(driver, 10).until(EC.presence_of_element_located((By.LINK_TEXT, "Logout")))

# 3️⃣ Click on Logout
driver.find_element(By.LINK_TEXT, "Logout").click()

# 4️⃣ Verify redirection to login/signup page
WebDriverWait(driver, 10).until(EC.url_contains("login"))
current_url = driver.current_url
```

```
assert "login" in current_url or "signup" in current_url, "❌ Logout failed"

print("✅ Test Passed: Successfully logged out and redirected to login page.")


# 5️⃣ Close browser

driver.quit()
```